



Neural Control and Coordination

The functions of the **organs/organ systems** in our body **must be coordinated to maintain homeostasis**.

- **Coordination** - the process through which **two or more organs interact and complement the functions of one another**.

Worked example - physical exercise. When we do physical exercises the **energy demand is increased** for maintaining an increased muscular activity, and the **supply of oxygen is also increased**. That increased oxygen supply necessitates an increase in the **rate of respiration, heart beat and increased blood flow via blood vessels**. When physical exercise is stopped, the activities of **nerves, lungs, heart and kidney** gradually return to their **normal conditions**. Thus the functions of **muscles, lungs, heart, blood vessels, kidney** and other organs are coordinated while performing physical exercises.

System	How it coordinates	Nature of the link
Neural system	Provides an organised network of point-to-point connections	For a quick coordination
Endocrine system	Provides chemical integration	Through hormones

In our body the **neural system and the endocrine system jointly** coordinate and integrate **all** the activities of the organs so that they function in a **synchronised fashion**.

① **[NOTE] Scope of this chapter.** You will learn about the **neural system of human and the mechanisms of neural coordination - the transmission of nerve impulse and impulse conduction across a synapse**.

18.1

Neural System

- **Neurons** - the neural system of **all** animals is composed of highly specialised cells called neurons, which can **detect, receive and transmit** different kinds of **stimuli**.

Animal group	Level of neural organisation
Lower invertebrates	Neural organisation is very simple .
Hydra	Composed of a network of neurons .
Insects	The neural system is better organised : a brain is present along with a number of ganglia and neural tissues .
Vertebrates	Have a more developed neural system.

18.2

Human Neural System

The human neural system is divided into **two parts**: (i) the **central neural system (CNS)** and (ii) the **peripheral neural system (PNS)**.

- **CNS** - includes the **brain** and the **spinal cord**, and is the **site of information processing and control**.
- **PNS** - comprises **all the nerves of the body associated with the CNS** (brain and spinal cord).

The nerve fibres of the PNS are of two types: (a) **afferent fibres** and (b) **efferent fibres**.

Fibre type	Direction of the impulse	What it carries
Afferent nerve fibres	From tissues/organs to the CNS	Sensory input arriving at the CNS
Efferent fibres	From the CNS to the concerned peripheral tissues/organs	Regulatory impulses

The PNS is divided into two divisions called the **somatic neural system** and the **autonomic neural system**.

Division of the PNS	Relays impulses from the CNS to
Somatic neural system	Skeletal muscles
Autonomic neural system	The involuntary organs and smooth muscles of the body

- The autonomic neural system is **further classified** into the **sympathetic neural system** and the **parasympathetic neural system**.
- **Visceral nervous system** - the part of the **peripheral nervous system** that comprises the whole complex of **nerves, fibres, ganglia and plexuses** by which impulses travel **from the central nervous system to the viscera** and **from the viscera to the central nervous system**.

① [NOTE] Exercise gap closed here (Exercise 10f: cranial nerves vs spinal nerves). NCERT's own text names **spinal and cranial nerves** only once, as the place where myelinated nerve fibres are found (18.3), and never defines either. Both are **PNS nerves** and they are told apart by **where they leave the CNS: cranial nerves arise from the brain, while spinal nerves arise from the spinal cord**. Each carries afferent fibres, efferent fibres, or both. Beyond this chapter's own sentences - kept because Exercise 10(f) demands the distinction (Rule 2).

18.3

Neuron as Structural and Functional Unit of Neural System

A **neuron** is a **microscopic** structure composed of **three major parts**, namely, **cell body, dendrites** and **axon** (Figure 18.1).

Part of the neuron	Structure	Function
Cell body	Contains cytoplasm with typical cell organelles and certain granular bodies called Nissl's granules	The metabolic centre of the neuron; impulses arrive here from the dendrites
Dendrites	Short fibres which branch repeatedly and project out of the cell body ; they also contain Nissl's granules	Transmit impulses towards the cell body
Axon	A long fibre , the distal end of which is branched ; each branch terminates as a bulb-like structure called a synaptic knob , which possesses synaptic vesicles containing chemicals called neurotransmitters	Transmits nerve impulses away from the cell body to a synapse or to a neuro-muscular junction

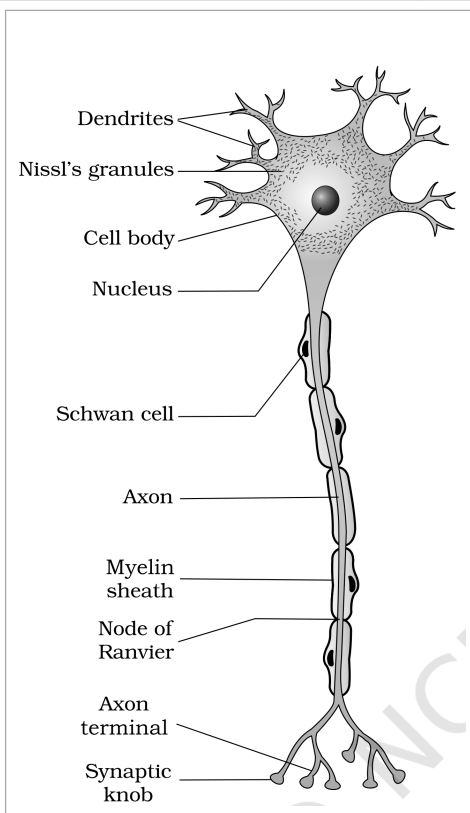


Fig. 18.1 - Structure of a neuron

① [NOTE] Read the plate (Figure 18.1 labels). The diagram marks the **Dendrites**, the **Nissl's granules** inside the **Cell body**, the **Nucleus** within that cell body, the **Schwan cell** (the plate's own spelling of the **Schwann cell**) wrapping the **Axon** as a **Myelin sheath**, a **Node of Ranvier** between two sheaths, and at the far end the **Axon terminal** ending in a **Synaptic knob**.

Based on the number of axon and dendrites, the neurons are divided into **three types**.

Type	Axon and dendrites	Where found
Multipolar	One axon and two or more dendrites	Cerebral cortex
Bipolar	One axon and one dendrite	Retina of eye
Unipolar	Cell body with one axon only	Found usually in the embryonic stage

There are **two types of axons**, namely, **myelinated** and **non-myelinated**.

Feature	Myelinated nerve fibre	Unmyelinated nerve fibre
Schwann cells	Enveloped with Schwann cells , which form a myelin sheath around the axon	Enclosed by a Schwann cell that does not form a myelin sheath around the axon
Gaps in the sheath	The gaps between two adjacent myelin sheaths are called nodes of Ranvier	No myelin sheath, hence no nodes of Ranvier
Where found	Spinal and cranial nerves	Commonly found in autonomous and the somatic neural systems

☆ [MEMORY AID - not in NCERT] **Neuron traffic direction: Dendrites Deliver in, Axons Away.** Both carry Nissl's granules only on the dendrite side of the cell body.

18.3.1

Generation and Conduction of Nerve Impulse

Neurons are **excitable cells** because their **membranes are in a polarised state**. Why is the membrane of a neuron polarised?

- **Different types of ion channels** are present on the neural membrane, and these ion channels are **selectively permeable to different ions**.
- **At rest** (a neuron **not conducting any impulse**) the axonal membrane is **comparatively more permeable to potassium ions (K^+)** and **nearly impermeable to sodium ions (Na^+)**.
- **Similarly**, the membrane is **impermeable to negatively charged proteins** present in the **axoplasm**.

Compartment	K^+	Na^+	Negatively charged proteins
Axoplasm (inside the axon)	High concentration	Low concentration	High - trapped inside
Fluid outside the axon	Low concentration	High concentration	Not stated by NCERT

These opposed concentrations **form a concentration gradient** across the membrane.

- **Sodium-potassium pump** - the ionic gradients across the resting membrane are maintained by the **active transport** of ions by this pump, which transports **3 Na^+ outwards for 2 K^+ into the cell**.

As a result, the **outer surface** of the axonal membrane possesses a **positive charge** while its **inner surface** becomes **negatively charged**, and the membrane is therefore **polarised**.

- **Resting potential** - the **electrical potential difference across the resting plasma membrane**.

Now the mechanism of generation of a nerve impulse and its conduction along an axon.

1

A **stimulus is applied at a site** (Figure 18.2, e.g., **point A**) on the polarised membrane; the membrane at **site A becomes freely permeable to Na^+** .

2

This leads to a **rapid influx of Na^+** , followed by the **reversal of the polarity at that site**: the **outer surface becomes negatively charged** and the **inner side becomes positively charged**. The polarity at site A is thus reversed and hence **depolarised**.

3

The electrical potential difference across the plasma membrane at site A is called the **action potential**, which is **in fact termed as a nerve impulse**.

4

At sites **immediately ahead** - the axon at **site B** - the membrane still has a **positive charge on the outer surface** and a **negative charge on its inner surface**.

5

As a result, a current flows on the inner surface from site A to site B; on the outer surface current flows from site B to site A (Figure 18.2), **completing the circuit of current flow**.

6

Hence the **polarity at the site is reversed** and an **action potential is generated at site B** - i.e. the impulse (action potential) generated at **site A arrives at site B**.

7

The sequence is repeated along the length of the axon and consequently the **impulse is conducted**.

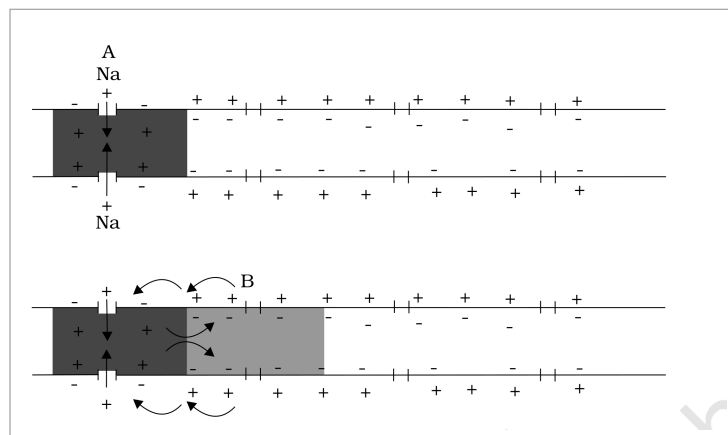


Fig. 18.2 - Diagrammatic representation of impulse conduction through an axon (at points A and B)

- ① **[NOTE] Read the plate (Figure 18.2 labels).** The plate marks the stimulated site **A** and the site immediately ahead, **B**, with rows of plus and minus charge marks on the two membrane surfaces and **Na** printed at each site where sodium (Na^+) rushes inwards. Reading the panels top to bottom shows the reversed polarity travelling from **A** to **B** along the axon.

Restoring the resting state. The stimulus-induced rise in permeability to Na^+ is extremely short-lived. It is quickly followed by a rise in permeability to K^+ . Within a fraction of a second, K^+ diffuses outside the membrane and restores the resting potential of the membrane at the site of excitation, and the fibre becomes once more responsive to further stimulation.

- ① **[NOTE] The impulse travels as a wave.** Taken end to end, the nerve impulse is **conducted along the axon membrane in the form of a wave of depolarisation and repolarisation**: depolarisation sweeps forward site by site while the sites just behind it are already being repolarised. (NCERT states this only in the chapter summary; it is folded in here, at the mechanism it describes.)

- ① **[NOTE] Exercise gap closed here (Exercise 10b: conduction in a myelinated vs an unmyelinated nerve fibre).** The chapter defines both fibre types (18.3) but never contrasts how they conduct. In an **unmyelinated** fibre the bare membrane is depolarised at **every successive point**, exactly as the A-to-B sequence above describes, so conduction is **continuous and slower**. In a **myelinated** fibre the **myelin sheath insulates the axon** and the membrane is exposed only at the **nodes of Ranvier**, so the impulse is regenerated **node to node - saltatory (jumping) conduction**, which is **faster** and uses less energy. Beyond this chapter's own sentences - kept because Exercise 10(b) demands the contrast (Rule 2).

A nerve impulse is **transmitted from one neuron to another through junctions called synapses.**

- **Synapse** - formed by the membranes of a **pre-synaptic neuron** and a **post-synaptic neuron**, which **may or may not be separated by a gap called the synaptic cleft.**

There are **two types of synapses**, namely, **electrical synapses** and **chemical synapses.**

Feature	Electrical synapse	Chemical synapse
Membranes	Membranes of pre- and post-synaptic neurons are in very close proximity	Membranes of the pre- and post-synaptic neurons are separated by a fluid-filled space called the synaptic cleft (Figure 18.3)
How the impulse crosses	Electrical current can flow directly from one neuron into the other across these synapses	Chemicals called neurotransmitters are involved in the transmission of impulses
Resemblance	Transmission is very similar to impulse conduction along a single axon	A fresh potential must be generated in the post-synaptic neuron
Speed	Always faster than that across a chemical synapse	Slower
Occurrence	Rare in our system	The usual synapse in our system

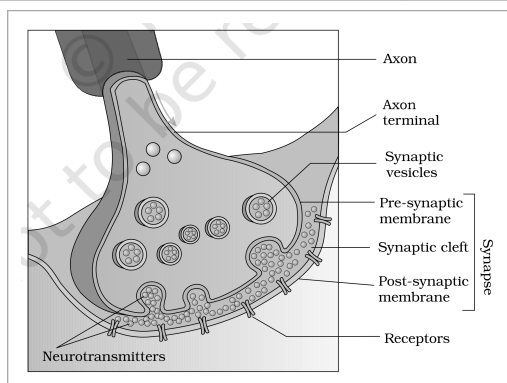


Fig. 18.3 - Diagram showing axon terminal and synapse

- ① **[NOTE]** Read the plate (Figure 18.3 labels). The plate follows the **Axon** into its **Axon terminal**, which holds the **Synaptic vesicles**; the enlarged inset of the **Synapse** shows the **Pre-synaptic membrane**, the **Synaptic cleft** and the **Post-synaptic membrane**, with **Neurotransmitters** crossing the cleft to the **Receptors** drawn on the post-synaptic membrane.

How does the pre-synaptic neuron transmit an impulse (action potential) across the synaptic cleft to the post-synaptic neuron?

- 1 The **axon terminals** contain **vesicles filled with neurotransmitters.**
- 2 When an **impulse (action potential)** arrives at the **axon terminal**, it **stimulates the movement of the synaptic vesicles towards the membrane.**
- 3 The vesicles **fuse with the plasma membrane** and **release their neurotransmitters in the synaptic cleft.**
- 4 The **released neurotransmitters bind to their specific receptors**, present on the **post-synaptic membrane.**
- 5 This **binding opens ion channels**, allowing the **entry of ions** which can **generate a new potential in the post-synaptic neuron.**
- 6 The **new potential developed may be either excitatory or inhibitory.**

The **brain** is the **central information processing organ** of our body, and acts as the '**command and control system**'.

What the brain does	Items NCERT lists
Controls	Voluntary movements; balance of the body; functioning of vital involuntary organs (e.g., lungs, heart, kidneys, etc.); thermoregulation; hunger and thirst; circadian (24-hour) rhythms of our body; activities of several endocrine glands ; and human behaviour
Site for processing of	Vision, hearing, speech, memory, intelligence, emotions and thoughts

Protection. The human brain is **well protected by the skull**. Inside the skull the brain is covered by **cranial meninges**.

Layer of the cranial meninges	Position
Dura mater	The outer layer
Arachnoid	A very thin middle layer
Pia mater	The inner layer, in contact with the brain tissue

The brain can be divided into **three major parts: (i) forebrain, (ii) midbrain and (iii) hindbrain** (Figure 18.4).

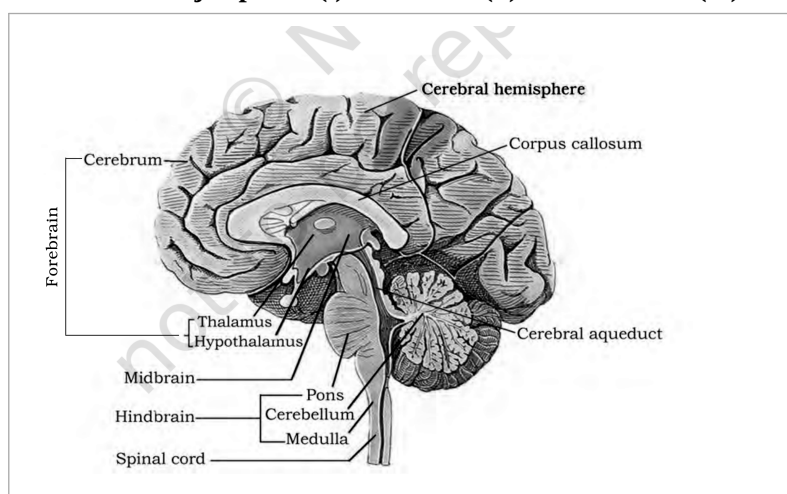


Fig. 18.4 - Diagram showing sagittal section of the human brain (NCERT prints 'sagital'; the conventional spelling is 'sagittal')

- ① **[NOTE]** Read the plate (Figure 18.4 labels). A bracket marks the **Forebrain**, inside which the plate names the **Cerebrum**, one **Cerebral hemisphere** of the pair, the **Corpus callosum** joining them, the **Thalamus** and the **Hypothalamus**. Below it are the **Midbrain**, with its **Cerebral aqueduct**, and the **Hindbrain** - **Pons, Cerebellum** and **Medulla** - which continues into the **Spinal cord**.

The forebrain consists of **cerebrum, thalamus** and **hypothalamus** (Figure 18.4).

- **Cerebrum** forms the major part of the human brain.
- A **deep cleft** divides the **cerebrum longitudinally** into two halves, which are termed the **left and right cerebral hemispheres**.
- **Corpus callosum** - the **tract of nerve fibres** that **connects the hemispheres**.
- **Cerebral cortex** - the **layer of cells** which **covers the cerebral hemisphere**; it is **thrown into prominent folds**.

Matter	What it is	Why it looks that way
Grey matter	The cerebral cortex is referred to as the grey matter	Due to its greyish appearance - the neuron cell bodies are concentrated here , giving the colour
White matter	Fibres of the tracts , covered with the myelin sheath , which constitute the inner part of the cerebral hemisphere	They give an opaque white appearance to the layer

The cerebral cortex contains **motor areas**, **sensory areas** and **large regions that are neither clearly sensory nor motor in function**.

- **Association areas** - those neither-sensory-nor-motor regions, responsible for **complex functions** like **intersensory associations**, **memory** and **communication**.

Forebrain part	Position	Function
Thalamus	The cerebrum wraps around it	A major coordinating centre for sensory and motor signaling
Hypothalamus	Lies at the base of the thalamus	Contains a number of centres which control body temperature, urge for eating and drinking ; also contains several groups of neurosecretory cells , which secrete hormones called hypothalamic hormones

- **Limbic lobe or limbic system** - the **inner parts of cerebral hemispheres** and a group of **associated deep structures** like **amygdala**, **hippocampus**, etc., form this complex structure.
- **Along with the hypothalamus**, the limbic system is involved in the **regulation of sexual behaviour**, the **expression of emotional reactions** (e.g., **excitement**, **pleasure**, **rage and fear**) and **motivation**.
- The limbic system is **also concerned with olfaction and autonomic responses**. (Stated only in the NCERT summary; folded in here with the rest of the limbic functions.)

① [NOTE] Exercise gap closed here (Exercise 9b: which part of our central neural system acts as a master clock?). The chapter states that the brain controls the **circadian (24-hour) rhythms** of our body (18.4) but never names the timekeeper. That role belongs to the **hypothalamus**, whose **suprachiasmatic nucleus** is the body's **master clock** - consistent with the hypothalamus already being the centre for body temperature, eating and drinking. The name of the nucleus is beyond this chapter's own sentences - kept because Exercise 9(b) demands it (Rule 2).

18.4.2

Midbrain

- The midbrain is **located between the thalamus/hypothalamus of the forebrain and the pons of the hindbrain**.
- A canal called the **cerebral aqueduct** passes through the midbrain (NCERT prints 'passess').
- The **dorsal portion** of the midbrain consists **mainly of four round swellings (lobes) called corpora quadrigemina**.
- **Function:** the midbrain **receives and integrates visual, tactile and auditory inputs**. (Stated only in the NCERT summary; the body gives the midbrain no function at all, so it is folded in here.)

18.4.3

Hindbrain

The hindbrain comprises **pons**, **cerebellum** and **medulla** (also called the **medulla oblongata**).

Hindbrain part	Structure	Function
Pons	Consists of fibre tracts	They interconnect different regions of the brain
Cerebellum	Has a very convoluted surface in order to provide additional space for many more neurons	Integrates information received from the semicircular canals of the ear and the auditory system (summary-only fact, folded in here)
Medulla	Connected to the spinal cord	Contains centres which control respiration, cardiovascular reflexes and gastric secretions

- **Brain stem** - **three major regions make up the brain stem: mid brain, pons and medulla oblongata**. The brain stem **forms the connections between the brain and spinal cord**.

☆ [MEMORY AID - not in NCERT] Hindbrain trio - 'PCM': Pons (bridges brain regions), Cerebellum (balance-and-hearing integrator, convoluted surface), Medulla (respiration, cardiovascular reflexes, gastric secretions). Drop the cerebellum and the remaining two, with the midbrain, are exactly the brain stem.

The **neural system coordinates and integrates functions** as well as **metabolic and homeostatic activities of all the organs**. **Neurons**, the **functional units** of the neural system, are **excitable cells** due to a **differential concentration gradient of ions across the membrane**. The electrical potential difference across the **resting** neural membrane is the '**resting potential**'; the nerve impulse is conducted along the axon membrane as a **wave of depolarisation and repolarisation**.

A **synapse** is formed by the membranes of a **pre-synaptic** and a **post-synaptic** neuron, which **may or may not** be separated by a gap called the **synaptic cleft**. Chemicals involved in the transmission of impulses at **chemical synapses** are called **neurotransmitters**.

The human neural system consists of **two parts: (i) the central neural system (CNS)** and **(ii) the peripheral neural system**. The **CNS consists of the brain and spinal cord** (the NCERT summary prints '**spiral cord**'). The brain divides into **(i) forebrain, (ii) midbrain and (iii) hindbrain**.

The **forebrain** consists of **cerebrum, thalamus and hypothalamus**. The **cerebrum is longitudinally divided into two halves** connected by the **corpus callosum**. The **hypothalamus, a very important part of the forebrain, controls body temperature, eating and drinking**. The **inner parts of the cerebral hemispheres plus associated deep structures form the limbic system, concerned with olfaction, autonomic responses, regulation of sexual behaviour, expression of emotional reactions and motivation**.

The **midbrain receives and integrates visual, tactile and auditory inputs**. The **hindbrain comprises pons, cerebellum and medulla**. The **cerebellum integrates information received from the semicircular canals of the ear and the auditory system**. The **medulla contains centres which control respiration, cardiovascular reflexes and gastric secretions**, and the **pons consists of fibre tracts that interconnect different regions of the brain**.

Three of NCERT's exercise questions lean on something this chapter never actually states. They are answered here, and **nothing in this appendix is NCERT body text** - each answer names the gap it fills. Every other exercise question is answered by the sections above.

9(b) Which part of our central neural system acts as a master clock?

- The **hypothalamus** - specifically its **suprachiasmatic nucleus** - which times the brain-controlled **circadian (24-hour) rhythms**.
- **The gap:** the chapter states that the brain controls circadian rhythms, but never names the structure that keeps the time.

10(b) Impulse conduction in a myelinated nerve fibre vs an unmyelinated nerve fibre.

- In an **unmyelinated** fibre the membrane is depolarised at **every successive point** along the axon, so conduction is **continuous** and **slower**.
- In a **myelinated** fibre the **myelin sheath insulates** the axon and leaves the membrane exposed only at the **nodes of Ranvier**, so the impulse is regenerated node to node and **leaps** the insulated stretches between them - **saltatory conduction**, which is **faster** and more energy-efficient.
- **The gap:** 18.1 defines both fibre types and 18.3 explains conduction along an axon, but the chapter never joins the two into a conduction comparison.

10(f) Cranial nerves vs spinal nerves.

- **Cranial nerves arise from the brain; spinal nerves arise from the spinal cord**. Both belong to the **PNS**, and either may carry **afferent** fibres, **efferent** fibres or both.
- **The gap:** the chapter names the two together only once - as the site where myelinated nerve fibres are found - and never distinguishes them.